EDA Insight Summary: Predicting Engine Life

Analytics Capstone Project (ANCP) | Deliverable D3 | Module M3: Exploratory Analysis | Group P | TCS (analytics partner) for GE Aerospace (client) | Data version data-v1.0 | Lead: Boomikaa P (BA05014); contributors: Ashish Kumar Shrivastava (BA05010), Bittu Kumar (BA05013) | Notebook: ANCP_TP_D3_eda.ipynb (+ HTML export)

1. Sprint 3 summary (Agile / Scrum view)

Sprint goal: turn the cleaned D2 data into evidence for the three layers of the project: **predictive** (RUL forecast), **prescriptive** (when to remove an engine) and **simulation** (cost against a fixed-interval schedule). **Business question (D1):** when should each in-service engine be taken off the aircraft? Hypotheses were written and fixed *before* any chart was drawn; each has a pass criterion and a link to this question.

ID	User story	Definition of done	Pts	Status
S3-1	See how engine life and RUL are distributed	Labelled lifetime charts; fixed-interval trade-off shown (Fig 1)	2	Done
S3-2	Know which sensors track wear	Ranked sensor correlations; trajectory vs RUL (Fig 2)	3	Done
S3-3	Compare segments (regimes, short vs long-lived engines)	Segment charts and numbers (Fig 3)	2	Done
S3-4	Repeat representativeness and bias checks on cleaned data	Train vs test tests on data-v1.0 (Fig 4)	2	Done
S3-5	Log insights and modelling implications	Insight log with evidence, so-what, confidence	2	Done
S3-6	Notebook reruns cleanly and is exported	Restart-and-run-all passes; HTML export made	1	Done; peer review pending

2. Hypotheses (stated before analysis)

ID	Hypothesis	Pass criterion (set in advance)	Business link
H1	Life varies so much that a fixed interval wastes life	Life CV >= 0.20 and shortest-life interval leaves >= 35% of mean life unused	Value of RUL rule vs fixed interval
H2	Several sensors drift toward failure	>= 5 sensors with |Spearman| >= 0.5 vs capped RUL	Predictive target is feasible
H3	Operating regime hides the FD004 signal	Regime explains >= 50% of raw variance; z-score helps >= 50% of sensors	Why D2 normalised by regime
H4	Degradation shows only late in life	Early change <= 25% of last-120-cycle change	When to act; RUL cap of 125
H5	Early readings do not predict total life	Max |rho| (first 20 cycles vs life) < 0.30	Decide in service, not at delivery
H6	Short-lived engines degrade faster	Spearman(late slope, life) <= -0.30	Need engine-specific trend features
H7	Test engines represent training engines	>= 90% sensors KS < 0.10 (cycles 1 to 30); life KS p > 0.05	Validity of all accuracy and cost claims
H8	FD004 is harder than FD001	Higher life CV and lower top-5 |rho|	Charter risk 4

3. Distributions and relationships

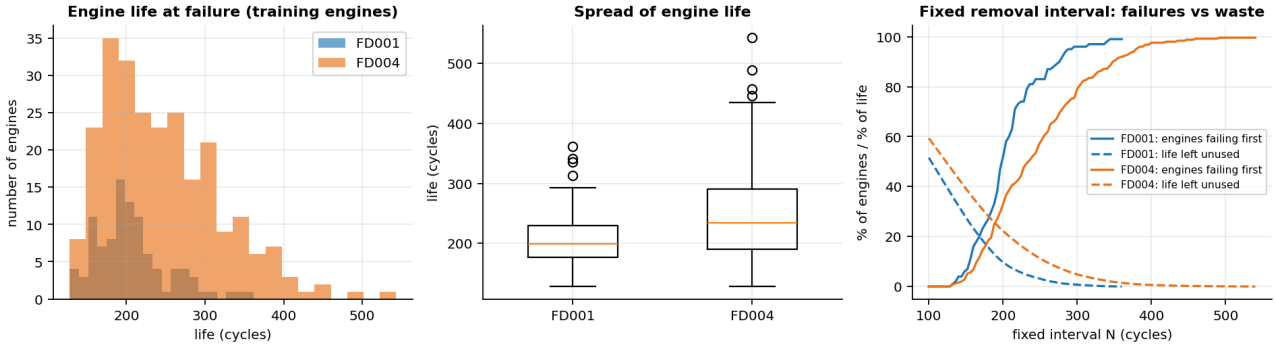


Figure 1. Engine life at failure (training engines). Right panel: for any fixed interval N, solid lines show the share of engines that fail first, dashed lines the share of life left unused.

Lives range from 128 to 362 cycles in FD001 (median 199) and 128 to 543 in FD004 (median 234). A fixed interval that no engine fails before (128 and 128 cycles) leaves 38% and 48% of life unused. Moving the interval to the 5th percentile still lets 4% and 5% of engines fail while wasting 29% and 39%. No single interval is good, which is the business case for a RUL-based rule.

Sensor ranking (Spearman vs capped RUL, regime-normalised): FD001 has 12 sensors with |rho| >= 0.5 (best s11 0.72, s4 0.70, s12 0.69); FD004 has 8 (best s11 0.71, s4 0.67, s17 0.63). Sensors s11 and s4 lead in both subsets; a few weak sensors (s6, s14 in FD001; s20, s21 in FD004) carry little signal.

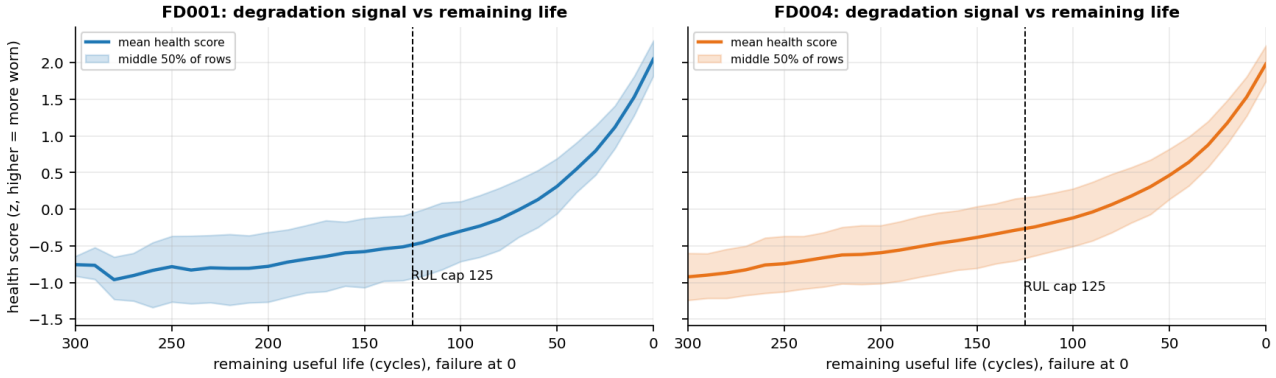


Figure 2. Health score (mean of the three strongest sensors, higher = more worn) against remaining life. The shaded band covers the middle 50% of rows; the dashed line marks the RUL cap of 125.

The score is nearly flat until about 125 cycles before failure and then rises steeply. Change before RUL 120 is only 15% (FD001) and 21% (FD004) of the change in the last 120 cycles, so early life tells us little about how far an engine is from failure.

4. Segments, representativeness and bias (repeated on the cleaned data)

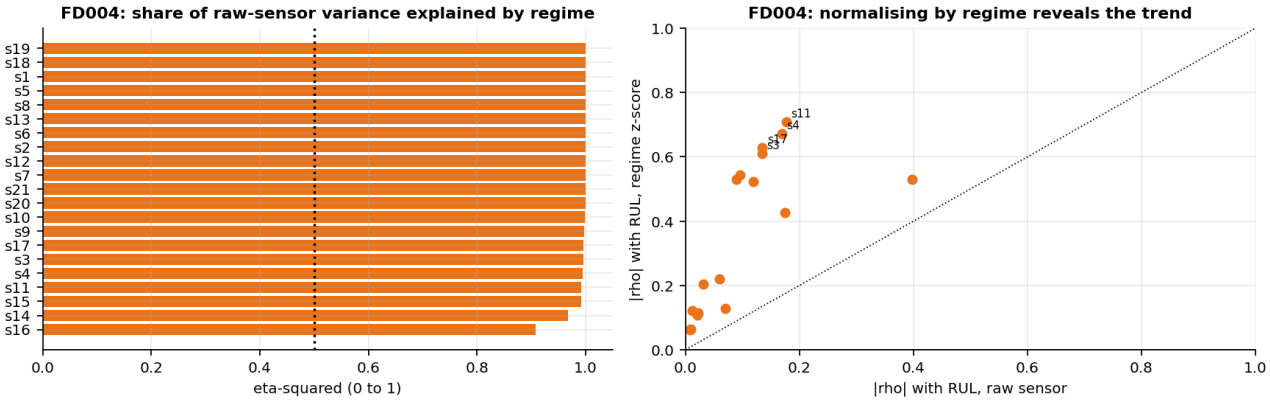


Figure 3. FD004 only. Left: share of each raw sensor's variance that is just the operating regime. Right: absolute correlation with RUL before (x) and after (y) regime normalisation.

Regimes. In FD004 the regime explains a median 100% of raw sensor variance, so raw readings mostly say how the engine is being flown, not how worn it is. After per-regime z-scoring the strongest sensor reaches |rho| 0.71 against 0.40 raw, and every sensor with a usable signal improves. The wear signal is then of similar size in all six regimes (late-life gap of 3.3 to 4.1 early-life standard deviations), so one model can serve all regimes.

Engine segments. In FD001 short-lived engines degrade faster near the end (rho -0.36); the effect is weak in FD004 (-0.15). Early readings do not predict life in FD001 (max |rho| 0.28) but do in FD004 (up to 0.54), which suggests initial wear differs between FD004 engines. This is correlation in simulated data, not proof of cause.

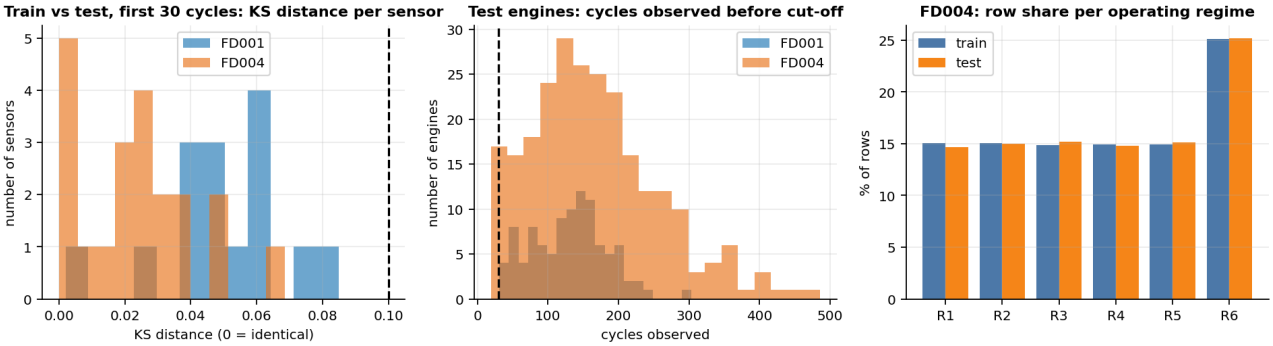


Figure 4. Train vs test on data-v1.0. Left: KS distance per sensor over the first 30 cycles (dashed = 0.10 limit). Middle: cycles observed per test engine (dashed = 30). Right: FD004 row share per regime.

Representativeness. Train and test look alike: every sensor has KS distance below 0.10 in the first 30 cycles (max 0.09 / 0.07), lifetime KS p-values are 0.99 and 0.86, and regime shares differ by at most 0.4 percentage points. **Bias and gaps:** 11 of 248 FD004 test engines are observed for fewer than 30 cycles (minimum 19); 39% and 49% of training rows have RUL above 125; long-lived engines are rare; and everything is simulated, with one engine class, no maintenance events and synthetic noise, so results are a proof of concept for GE.

5. Hypothesis scorecard and insight log

ID	Hypothesis	FD001	FD004	Evidence (FD001 / FD004)
H1	Life varies too much for one fixed interval	Supported	Supported	CV 0.22 / 0.30; unused at no-failure interval 38% / 48%
H2	Sensors drift toward failure	Supported	Supported	sensors with |rho|>=0.5: 12 / 8; best |rho| 0.72 / 0.71
H3	Regime hides the FD004 signal	n/a	Supported	median eta2 1.00; z beats raw for 100% of sensors
H4	Degradation visible only late in life	Supported	Supported	early/late change ratio 0.15 / 0.21
H5	Early readings do not predict life	Supported	Not supported	max |rho| 0.28 / 0.54
H6	Short-lived engines degrade faster	Supported	Not supported	rho -0.36 / -0.15
H7	Test is representative of train	Supported	Supported	sensors with KS<0.10: 100% / 100%; life KS p 0.99 / 0.86
H8	FD004 is harder than FD001	n/a	Supported	CV 0.22 vs 0.30; top-5 |rho| 0.70 vs 0.63

Two hypotheses are not supported in FD004 (H5 and H6). That is a finding, not a failure: it tells us FD004 engines carry more initial-state information and less consistent degradation speed than FD001 engines.

ID	Insight	Evidence (FD001 / FD004)	So-what for the decision	Confidence
I1	Engine life varies widely, so any fixed interval either wastes life or lets engines fail	Life CV 0.22 / 0.30; min 128 vs median 199 (FD001), min 128 vs median 234 (FD004). Interval = shortest life leaves 38% / 48% unused; interval at 5th percentile still fails 4% / 5% and wastes 29% / 39% (notebook Fig 1)	Strong business case for a RUL-based removal rule; the D6 simulation baseline should be this failure-vs-waste curve	High (data); Medium for real fleets
I2	A handful of sensors track wear: s11 and s4 lead in both subsets	12 (FD001) and 8 (FD004) sensors with |rho| >= 0.5; best |rho| 0.72 / 0.71 (notebook Fig 2)	RUL is predictable from sensors; start features with the top sensors, prune the rest in D4	High
I3	Degradation shows up mainly in the last ~120 cycles	Signal change before RUL 120 is only 15% / 21% of the change in the last 120 cycles (notebook Fig 3)	Supports RUL cap near 125 and a rule that acts on predicted RUL (threshold), not on age	Medium-High; cap value tuned in D4
I4	In FD004 the operating regime, not wear, dominates raw sensor values	Regime explains median 100% of raw variance; regime z-scores improve |rho| for 100% of sensors (notebook Fig 4)	Never feed raw FD004 sensors to a model; use the _z features or regime-aware models	High
I5	Late-life degradation speed differs by engine, and early readings carry information in FD004	Slope vs life rho -0.36 (FD001) / -0.15 (FD004); early-reading vs life |rho| up to 0.28 / 0.54 (notebook Fig 6)	Add rolling-trend and initial-state features per engine; test them in a D4 ablation (not causal proof)	Medium
I6	Train and test engines look alike, but some test histories are very short	All sensors KS < 0.10 in the first 30 cycles; life KS p 0.99 / 0.86; regime share gap <= 0.4 pts; 11 FD004 test engines have < 30 cycles (min 19) (notebook Fig 8)	Evaluation is fair within the simulator; use windows <= 19 cycles or padding and report short-history error separately	High (split); Low (real fleet)
I7	FD004 is harder than FD001	Life CV 0.30 vs 0.22; top-5 median |rho| 0.63 vs 0.70	Tune on FD001 first, report FD004 as the harder case (charter risk 4)	Medium-High

6. Summary: the five insights that matter most and how they shape modelling

#	What we learned	How it shapes modelling (D4 onward)
1	Fixed intervals are costly: life CV 0.22 / 0.30; waste of 38% / 48% even at the safest interval	Baseline = mean life minus current cycle. The D6 simulation compares the RUL rule with the Figure 1 failure-vs-waste curve, using Weibull lives from D2 and a cost sweep (failure : early removal 3 to 30).
2	Few sensors carry the signal (s11, s4 first in both subsets, |rho| about 0.7)	Start with the strongest sensors; prune weak and redundant ones. Add rolling mean and slope over a past window of at most 19 cycles.
3	Wear shows only in the last ~120 cycles	Train on RUL capped at 125 but score on uncapped RUL; the prescriptive rule triggers on predicted RUL below a threshold, tuned in D6. Late errors must cost more than early ones.
4	FD004 raw sensors mostly reflect the regime; FD004 is harder	Use the regime-normalised features in every model; report FD001 first and FD004 as the harder case, with error by regime.
5	Train and test match, but 11 FD004 test engines have very short histories	Split and cross-validate by engine only; recompute z-statistics inside each training fold; report short-history error separately.

7. Sprint review, retrospective (draft) and next sprint

Sprint review: all five analysis stories are done and the notebook was executed top to bottom in a fresh kernel; peer review by Ashish and Bittu is pending. **Retrospective (draft; edit after the team meeting):** hypotheses written first kept the analysis honest and surfaced two useful negative results; next time, agree chart styles before plotting. **Actions:** tag the repository *eda-v1*; update the Kanban board and burndown.

Next sprint story	Layer	Due
S4-1 Baseline model and metric code (RMSE, NASA asymmetric score), engine-wise split	Predictive	2 Nov
S4-2 Candidate models on FD001 then FD004; MLflow experiment log	Predictive	2 Nov
S4-3 Prototype repair-timing threshold on predicted RUL	Prescriptive	5 to 6 Nov
S4-4 Cost simulation skeleton (Weibull lives, failure : early cost sweep)	Simulation	6 Nov

Files: ANCP_TP_D3_eda.ipynb, ANCP_TP_D3_eda.html, ANCP_TP_D3_summary.pdf. Rename TP to your team number.